

Part I. Diagnostic confirmation of suspected EB

Diagnosis of EB

Providing a precise diagnosis is the first step in approaching a patient suspected of having epidermolysis bullosa (EB) and will yield invaluable prognostic information. The diagnosis of EB is based on a combination of clinical features, family history, and laboratory findings. Laboratory techniques include immunofluorescence mapping (IFM), transmission electron microscopy (TEM), and molecular genetic testing. The methods for laboratory diagnosis of EB have been described in detail previously.

1. Clinical manifestations

- Recurrent blisters and/or erosions of the skin and/or oral mucosa since birth, or starting later in life, induced by mild physical trauma.
- Congenital skin loss.

Differential diagnosis should be excluded based on clinical features and laboratory testing (e.g., microbiologic swabs, histopathology, immunodiagnosics):

- **Neonates:** infections (e.g., staphylococcal scalded skin syndrome, multifocal herpes simplex), other genetic disorders with skin fragility or blistering (e.g., epidermolytic ichthyosis), diffuse cutaneous mastocytosis, autoimmune blistering disorders (extremely rare).
- **Infancy and childhood:** infections, bullous mastocytosis, other genetic disorders with skin fragility, dermatitis artefacta, autoimmune blistering disorders (rare).

- **Adults:** infections, bullous/erosive adverse drug reactions, autoimmune blistering disorders, dermatitis artefacta.

2. Samples for EB laboratory diagnosis (informed consent required)

Whenever possible, laboratory diagnosis should be performed in a specialized EB centre.

- **Genetic testing:** An ethylenediaminetetraacetic acid (EDTA) blood sample (2–5 mL), or alternatively saliva or buccal smear, should be obtained from the patient and, where applicable, both parents.
- **Skin biopsy:** A 4- to 6-mm punch or superficial shave biopsy from a fresh blister (less than 12 hours old) or induced blister, including perilesional skin, is required. Blisters can be induced by gently rotating a pencil eraser on non-blistered skin.
- The biopsy can be stored or immediately shipped in sterile saline, Michel's medium, or frozen for immunofluorescence mapping (IFM). Some laboratories perform immunohistochemistry on formalin-fixed and paraffin-embedded tissue.
- For transmission electron microscopy (TEM), the biopsy must be placed in appropriate fixative. The diagnostic laboratory should be consulted in advance for specific requirements.
- **Topical anaesthetic creams** should be avoided, as they may cause artificial epidermal cleavage.
- When inducing a blister, gently rub intact or perilesional skin.
- In cases of extreme skin fragility, care must be taken to avoid epidermal detachment or rupture of the blister roof during biopsy.

3. Specific tests for EB laboratory diagnosis

- **Genetic testing** is the gold standard, providing a definitive diagnosis and classification of the major EB type and, in many cases, the subtype. It also enables genetic counselling and DNA-based prenatal testing.

- A next-generation sequencing (NGS) EB-gene panel is the preferred method, as it includes all known EB-associated genes and ensures comprehensive coverage.
- Sanger sequencing may be used when the candidate gene is known and small, or to confirm familial or recurrent variants.
- The vast majority of EB cases can be genetically characterized using these approaches.